

Lecture 25

Large Language Model (LLM) Multi-Agent Systems

COMP532 – Machine Learning and Bioinspired Optimisation

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Today's Questions

At the end of this lecture, you should be able to answer:

1. Why use multiple LLM agents instead of a single model?
2. What kinds of interaction patterns emerge in LLM multi-agent systems?
3. What new failure modes arise?
4. How can we analyse these systems using game theory and dynamics?

From Single-Agent to Multi-Agent LLM Systems

Traditional LLM use:

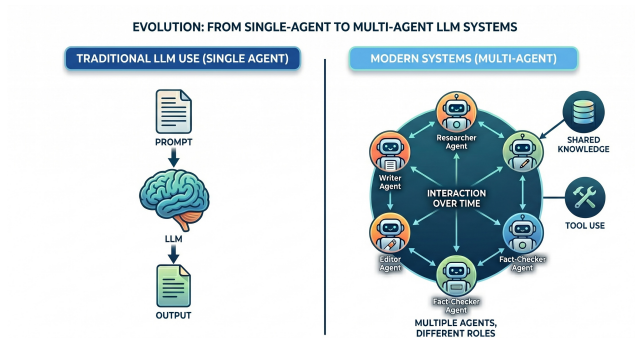
- ▶ one model
- ▶ one prompt
- ▶ one output

Modern systems:

- ▶ multiple agents
- ▶ different roles
- ▶ interaction over time

Shift

From **static generation** to **interactive systems**.



Why Multiple Agents?

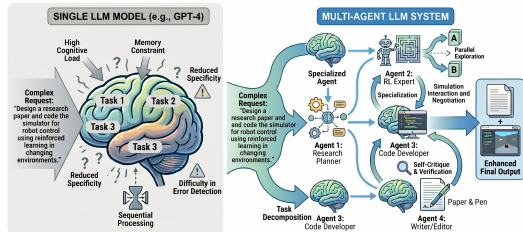
Why not just use a stronger single model?

Multi-agent systems aim to:

- ▶ decompose complex tasks
- ▶ enable specialization
- ▶ allow self-critique and verification
- ▶ support parallel exploration
- ▶ simulate interaction and negotiation

Key idea

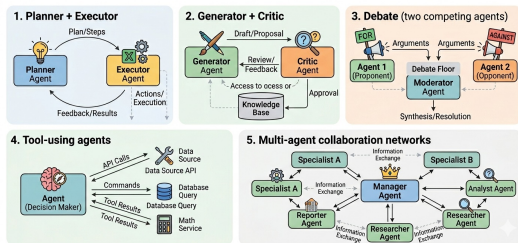
Multiple agents introduce **structure and interaction**.



Typical Architectures

Common patterns include:

- ▶ Planner + Executor
- ▶ Generator + Critic
- ▶ Debate (two competing agents)
- ▶ Tool-using agents
- ▶ Multi-agent collaboration networks



Observation

These architectures implicitly define a **game or interaction structure**.

Projects

CrewAI, AutoGen, LangGraph / LangChain multi-agent workflows and so on.

Example: Reflection / Self-Refinement

- ▶ Agent generates an initial answer
- ▶ Then critiques or reflects on it
- ▶ Then revises its own output
- ▶ This process repeats over multiple steps

Example prompt

You are solving a problem.

1. Write your initial answer.
2. Reflect on your answer: - What might be incorrect? - What assumptions are weak?
3. Improve your answer based on reflection.

Interpretation

This creates a **feedback loop**, where outputs are iteratively improved.

Interaction Creates Dynamics

Once agents interact, we get:

- ▶ feedback loops
- ▶ adaptation over time
- ▶ dependence on previous outputs
- ▶ path-dependent behaviour

Example: Reflection

$$y^{(0)} \rightarrow y^{(1)} \rightarrow y^{(2)}$$

- ▶ the first answer affects the critique
- ▶ the critique affects the revision
- ▶ later outputs depend on earlier ones

Connection

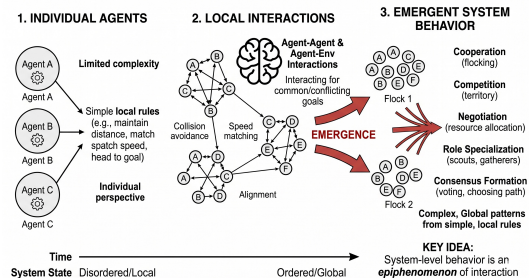
This is exactly the kind of behaviour we studied as:

- ▶ evolutionary dynamics
- ▶ learning dynamics

Emergent Behaviour

Multi-agent systems can exhibit behaviours not present in single agents:

- ▶ cooperation
- ▶ competition
- ▶ negotiation
- ▶ role specialization
- ▶ consensus formation



Key idea

System-level behaviour emerges from **interaction**.

A Game-Theoretic View

We can interpret LLM agents as players:

- ▶ actions: generated outputs
- ▶ payoffs: evaluation scores / task success
- ▶ strategies: prompting + reasoning patterns

Perspective

LLM systems can be analysed as **games between agents**.

A Dynamical Systems View

From Previous Lectures:

- ▶ policies evolve over time
- ▶ updates depend on feedback
- ▶ interaction induces trajectories

In LLM agents:

- ▶ outputs influence future prompts
- ▶ agents adapt implicitly via context

Insight

LLM multi-agent systems are **dynamical systems**.

Example: Debate Systems

Two agents argue for different answers.

Example

Question: What is the capital of Australia?

- ▶ **Agent 1:** I argue for Sydney, because it is the largest and best-known city.
- ▶ **Agent 2:** I disagree. Canberra is the capital, even though it is less famous.
- ▶ **Judge:** Canberra is correct. The debate exposes the error in Agent 1's reasoning.

This can:

- ▶ improve reasoning quality
- ▶ expose errors
- ▶ reduce hallucination

But also:

- ▶ lead to strategic behaviour
- ▶ create persuasive but incorrect arguments

Example: Coordination Problems

Agents may need to:

- ▶ agree on a plan
- ▶ divide tasks
- ▶ share information

Example: report-writing agents

Task: Write a short report on climate policy.

- ▶ **Agent 1 output:** I will summarise policy goals and key regulations.
- ▶ **Agent 2 output:** I will collect recent examples and evidence.
- ▶ **Shared result:** The final report covers both policy structure and real-world cases.

Possible outcomes:

- ▶ successful coordination
- ▶ inefficient coordination
- ▶ failure to agree

Stability and Convergence

Key questions:

- ▶ Do agents converge to consistent behaviour?
- ▶ Do interactions stabilize or oscillate?
- ▶ Are outcomes robust to perturbations?

Connection

These are exactly the questions studied in:

- ▶ EGT (stability)
- ▶ RL dynamics (convergence)

Why This Matters

Understanding LLM multi-agent systems requires:

- ▶ reasoning about interaction
- ▶ analysing dynamics
- ▶ studying emergent behaviour
- ▶ designing stable systems

Main message

We need tools from **game theory, learning, and dynamics**.

A Mental Model

- ▶ EGT: how behaviours evolve in populations
- ▶ RL: how policies change through feedback
- ▶ LLM MAS: how interacting agents produce system-level behaviour

Modern AI systems are fundamentally multi-agent and dynamic.

Summary

1. LLM systems are increasingly multi-agent
2. interaction introduces dynamics and feedback
3. emergent behaviour can arise
4. game theory and dynamics provide analysis tools